

A TIME SERIES PROJECT

ANALYSIS AND PREDICTION OF ELECTRICITY PRODUCTION BASED ON PAST 33 YEARS IN INDIA

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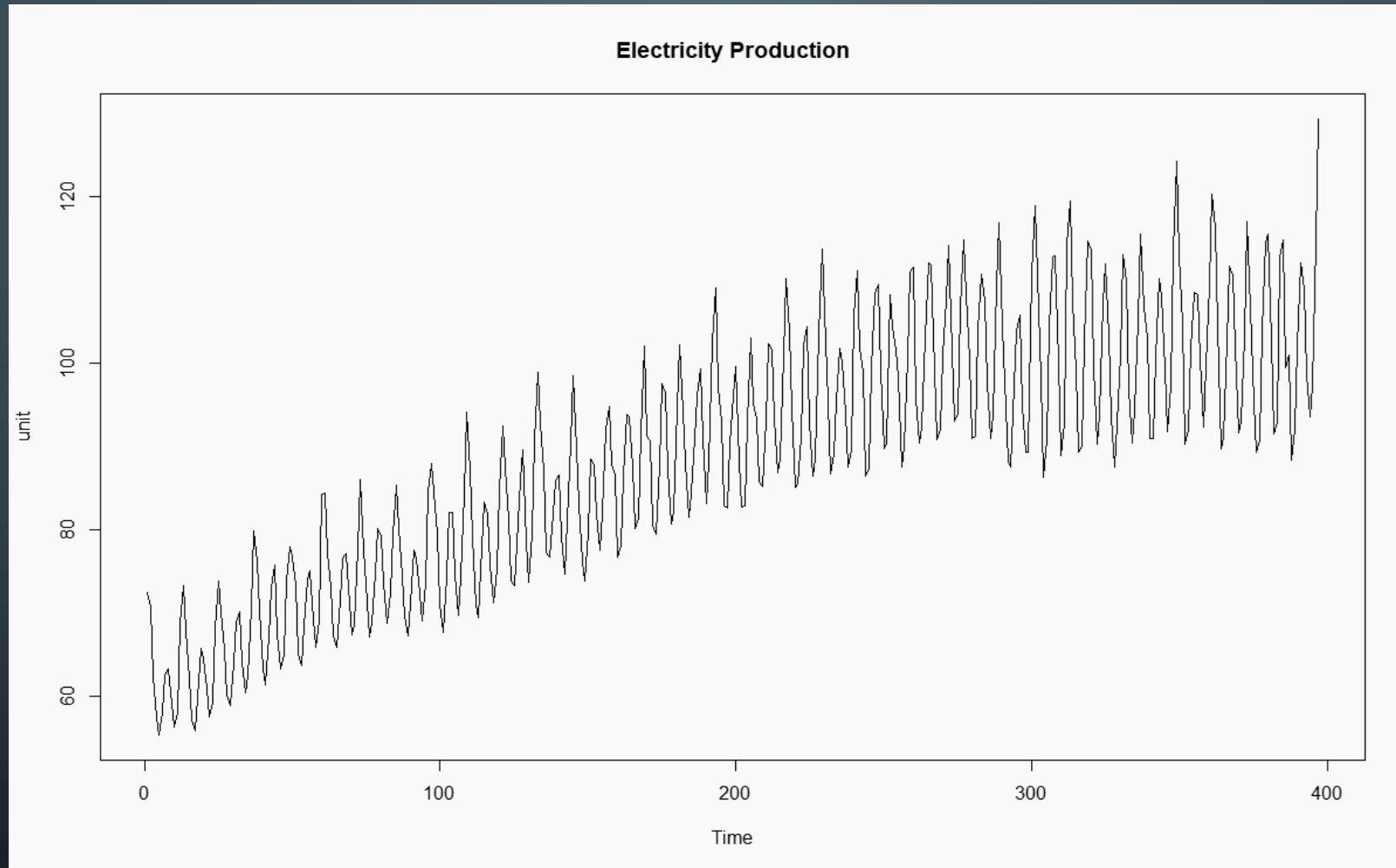
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DATA DESCRIPTION

- We have taken a monthly production data of electricity in India with 397 data points from January 1985 to January 2018.
- Dates(t) are in (MM/DD/YYYY) format
- Electric Production(Yt) is in some Unit

Date (MM/DD/YYYY)	Electric Production (in unit)
1/1/1985	72.5052
2/1/1985	70.672
3/1/1985	62.4502
4/1/1985	57.4714
5/1/1985	55.3151
6/1/1985	58.0904
7/1/1985	62.6202
8/1/1985	63.2485
9/1/1985	60.5846
10/1/1985	56.3154
11/1/1985	58.0005
12/1/1985	68.7145
1/1/1986	73.3057
2/1/1986	67.9869
3/1/1986	62.2221
4/1/1986	57.0329
... ..	

RAW DATA PLOT



We can say that, there may have existence of trend and seasonality

OUR OBJECTIVES

1. Testing the presence of trend.
2. Testing the presence of seasonality.
3. Determination the period of seasonality.
4. If trend and seasonality exist, then estimation and elimination of them.
5. Checking the randomness of the final de-seasonalised and de-trended data.
6. Finding out the probability model that well fits the random data and estimation of its parameters.
7. Forecasting.

1. TESTING THE PRESENCE OF TREND

Relative Ordering Test :

H_0 : Trend is absent Vs H_1 : Trend is present

- Test statistic : $Z = \frac{T}{\sqrt{\frac{2(2n+5)}{9n(n-1)}}} \sim AN(0, 1)$ under H_0
- Rule : Reject H_0 if $|Z_{obs}| > \tau_{\alpha/2}$
- We found $|Z_{obs}| = 19.1257 > \tau_{0.025} = 1.96$. Hence conclude that trend is present.

2. TESTING THE PRESENCE OF SEASONALITY

Friedman's test :

H_0 : no seasonality present vs H_1 : Seasonality is present

- Test statistic:

$$X = 12 \sum_{i=1}^{12} \frac{(M_i - \frac{c(12+1)}{2})^2}{c \times 12 \times (12+1)} \sim A\chi_{12-1}^2 \text{ under } H_0$$

- Rule: Reject H_0 if $X_{obs} > \chi_{\alpha,11}^2$
- We found $X_{obs} = 2882.333 > \chi_{0.05,11}^2 = 19.675$, so we conclude that seasonality is present in the process

3. DETERMINATION OF PERIOD OF SEASONALITY

We consider the process $\nabla_d Y_t$ by using lag delta operator with $d=12$,

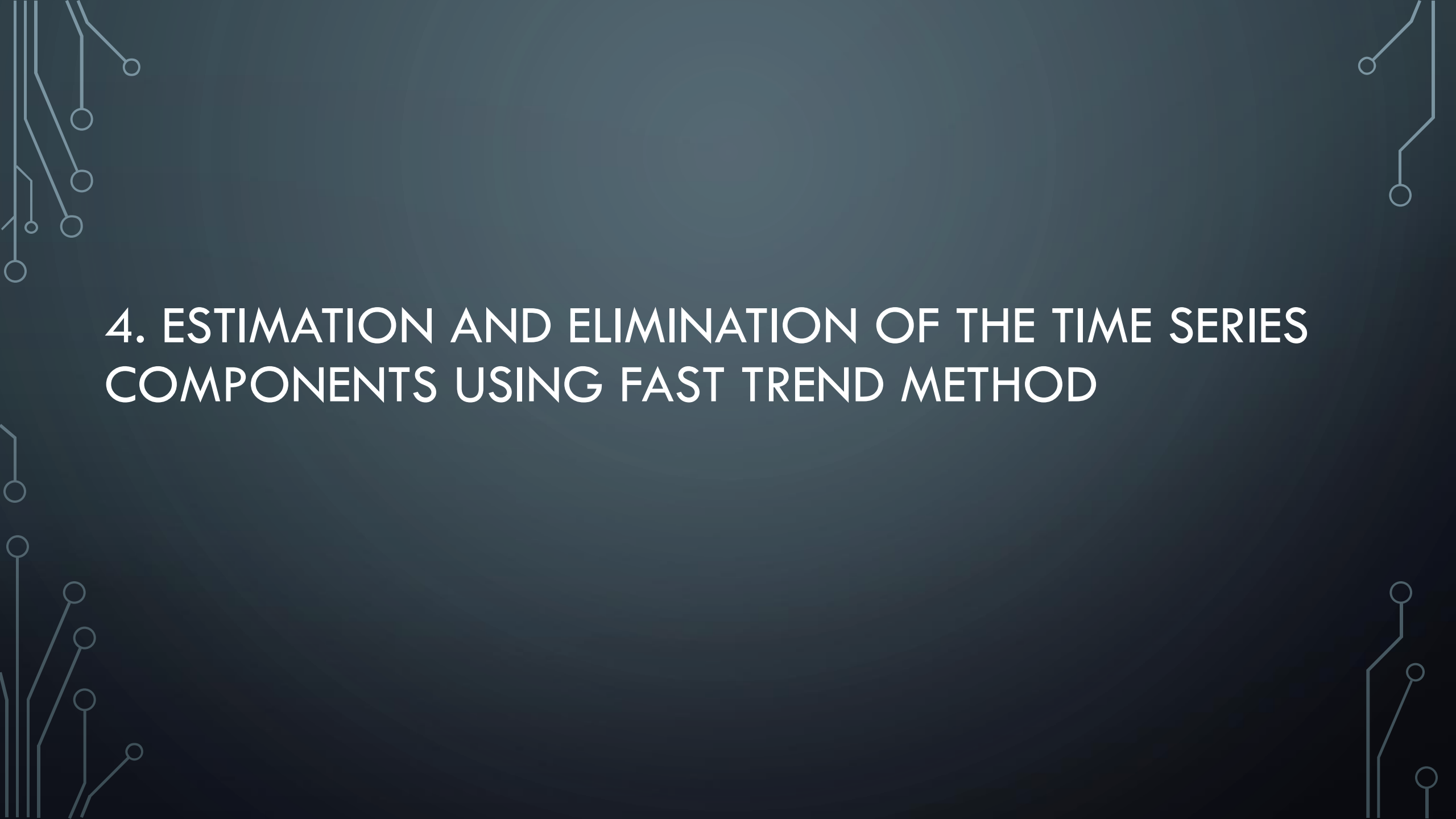
Then performed following test on $\nabla_{12} Y_t$

Turning point test :

H_0 : series is purely random Vs H_1 : not random

- Test statistic: $Z = \frac{P - \frac{2}{3}(n-2)}{\sqrt{\frac{16n-29}{90}}} \sim AN(0, 1) \text{ under } H_0$
- Rule : Reject H_0 if $|Z_{obs}| > \tau_{\alpha/2}$

We found $|Z_{obs}| = 0.70 < \tau_{0.025}$ hence we can conclude that the series $\nabla_{12} Y_t$ is random.
Which means the period of seasonality is 12.

The background is a dark blue gradient. In the corners, there are decorative white line art elements resembling circuit boards or neural network connections. These elements consist of thin lines that branch out and terminate in small circles, creating a symmetrical, abstract pattern in each corner.

4. ESTIMATION AND ELIMINATION OF THE TIME SERIES COMPONENTS USING FAST TREND METHOD

- Model: $Y_t = m_t + S_t + I_t$
- m_t : Trend component, S_t : Seasonal component, I_t : irregular component
- Model assumptions are $\{I_t\}$ is a covariance stationary process with mean 0 and variance σ^2 .
- Period $d=12$ (even)

A. ROUGH ESTIMATION OF TREND

We first obtained a rough estimate of trend by the method of moving average.

- The 12 point MA estimate of trend :

$$\hat{m}_t = \frac{1}{12} \left(\frac{1}{2}Y_{t-6} + Y_{t-5} + \dots + Y_t + \dots + Y_{t+5} + \frac{1}{2}Y_{t+6} \right), 6 < t < 391$$

- We have estimated the first 6 and the last 6 values using the endpoint padding method.

B. ESTIMATION OF SEASONAL COMPONENT

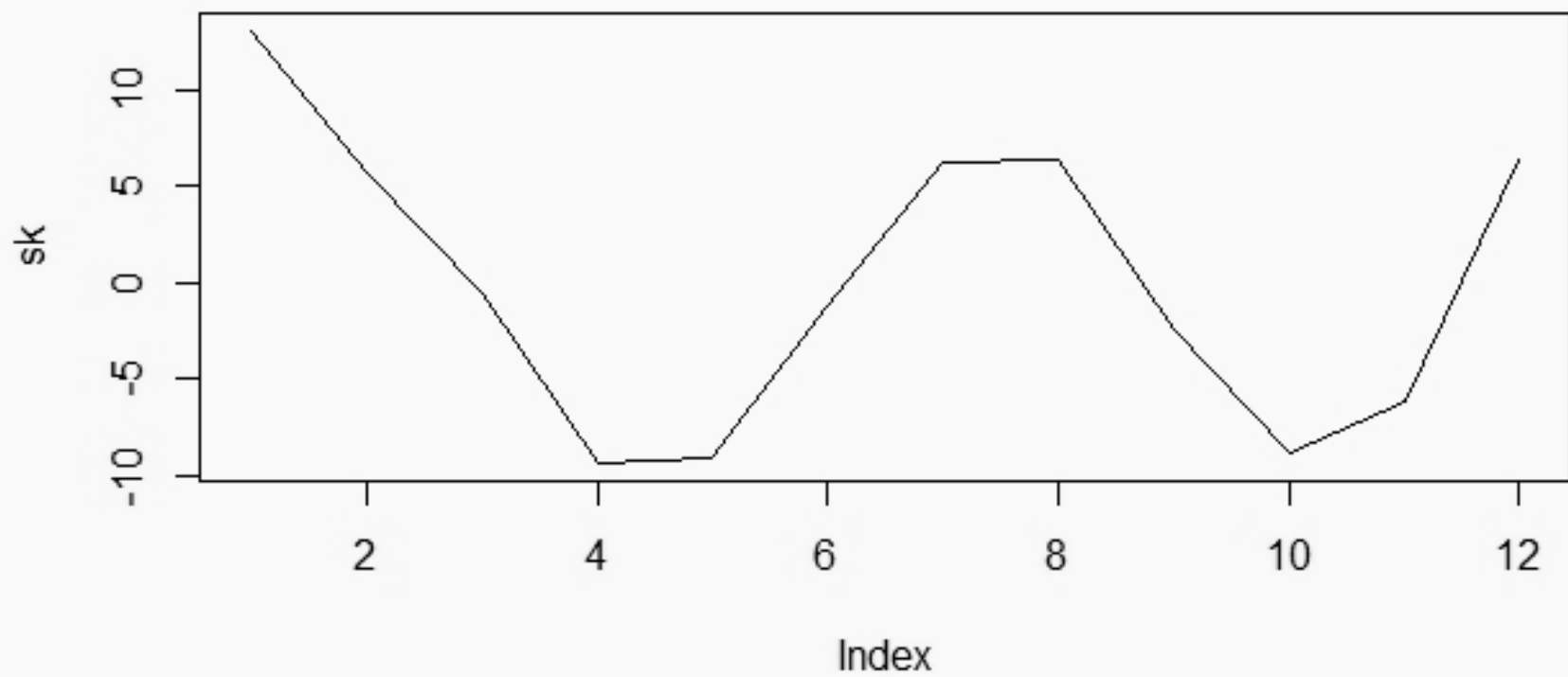
- For each $k = 1, 2, \dots, 12$ we define,

$$W_k = \frac{1}{33} \sum_{j=1}^{33} (Y_{k+(j-1)d} - \hat{m}_{k+(j-1)d})$$

- So 12 point estimates of seasonal components are :

$$\hat{S}_k = W_k - \frac{1}{12} \sum_{k=1}^{12} W_k \quad k = 1, 2, \dots, 12$$

Plot of estimated seasonal Component



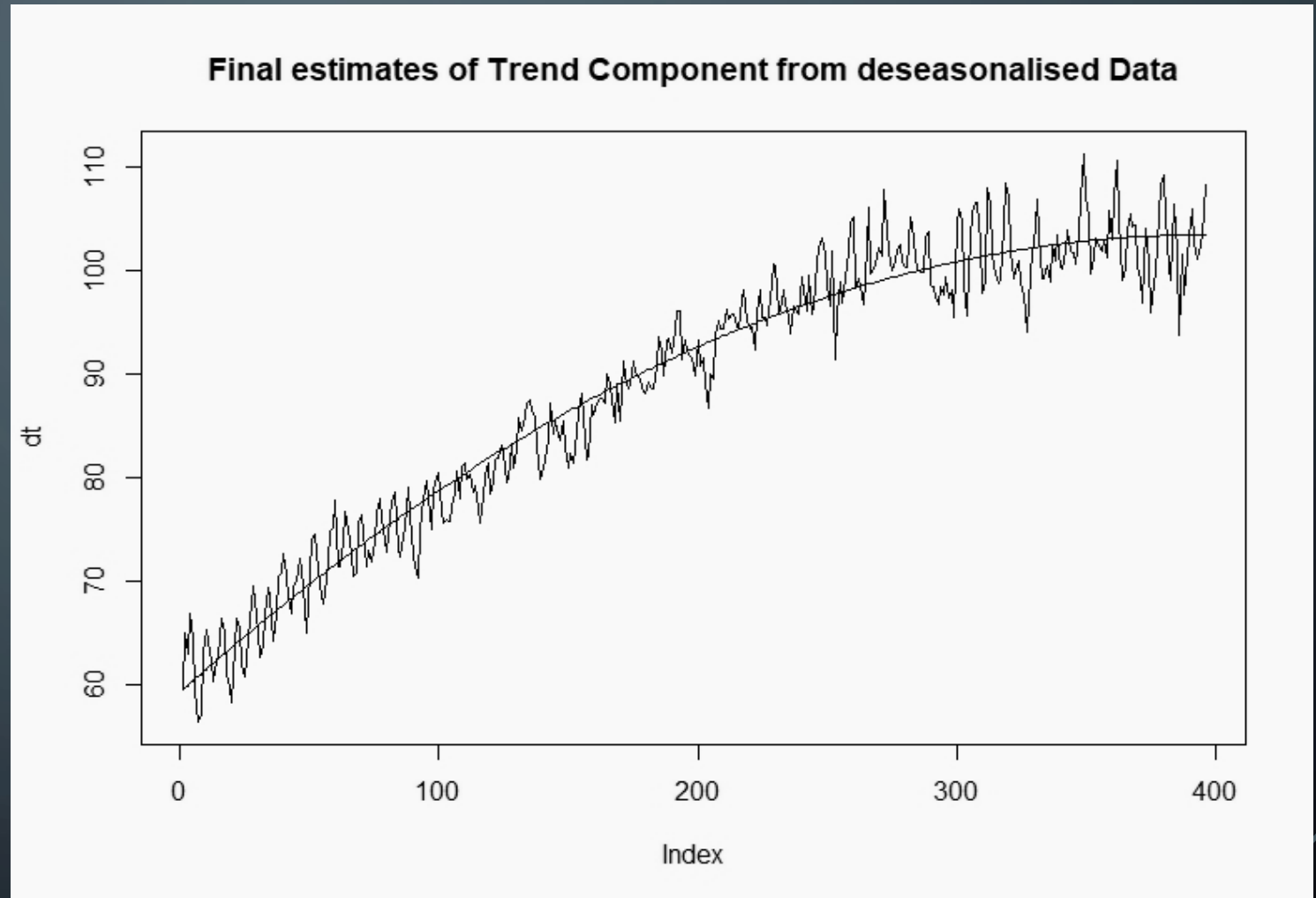
C. RE-ESTIMATION OF TREND AFTER DE-SEASONALISATION

- De-seasonalised data

$$d_t = Y_t - \hat{S}_t$$

- Then we re-estimated the trend component using polynomial trend
- The estimated trend for the d_t series is given by,

$$\hat{\hat{m}}_t = 59.1781 + 0.2235t - 0.000282t^2$$

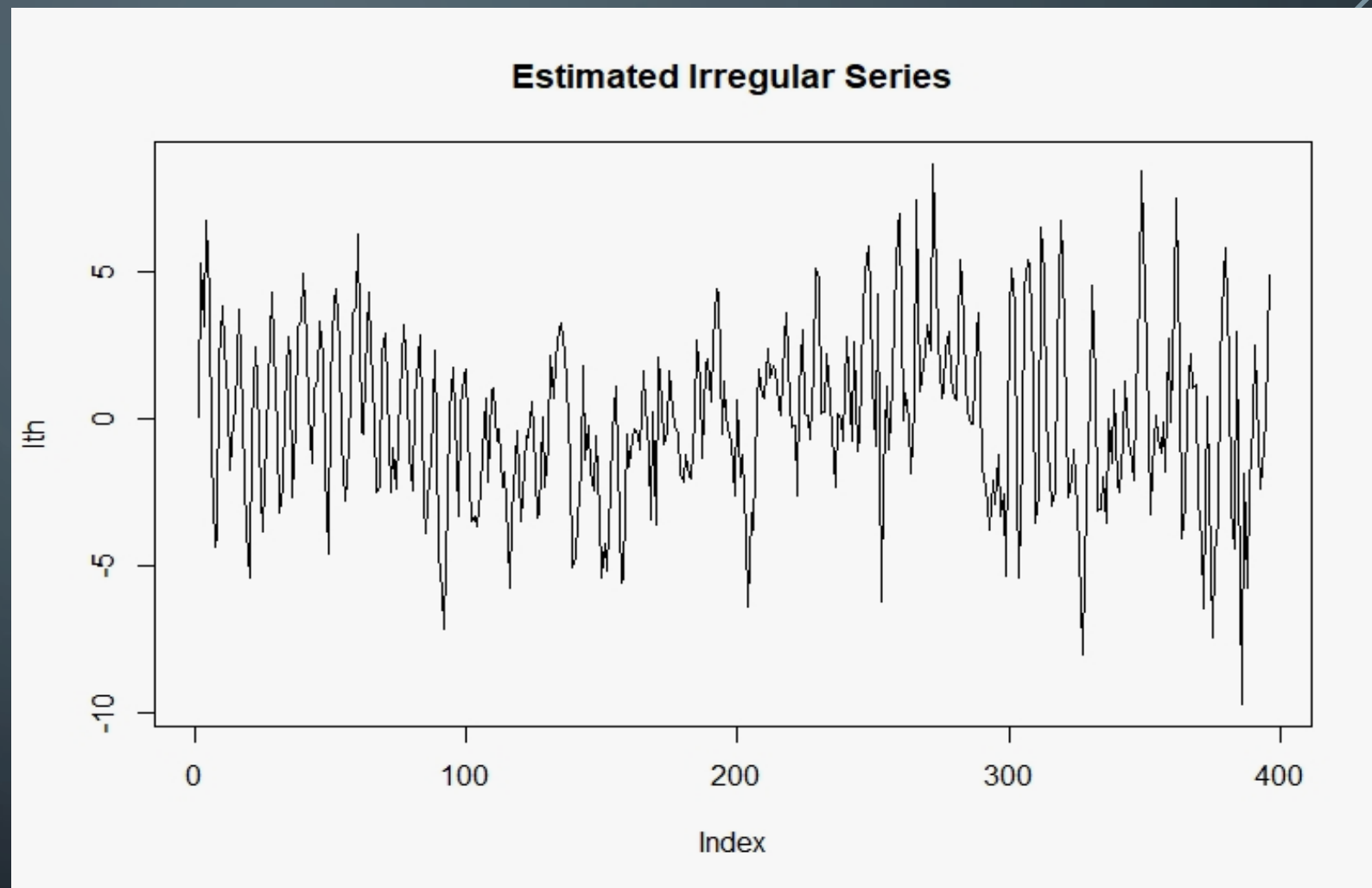


D. ESTIMATION OF THE IRREGULAR COMPONENT

- Our estimated irregular series is given by,

$$\hat{I}_t = Y_t - \hat{\hat{m}}_t - \hat{S}_t$$

- It is looking like random.
But let's check it by turning point test



We found $|Z_{obs}| = 1.01 < \tau_{0.025} = 1.96$. So the randomness is significant

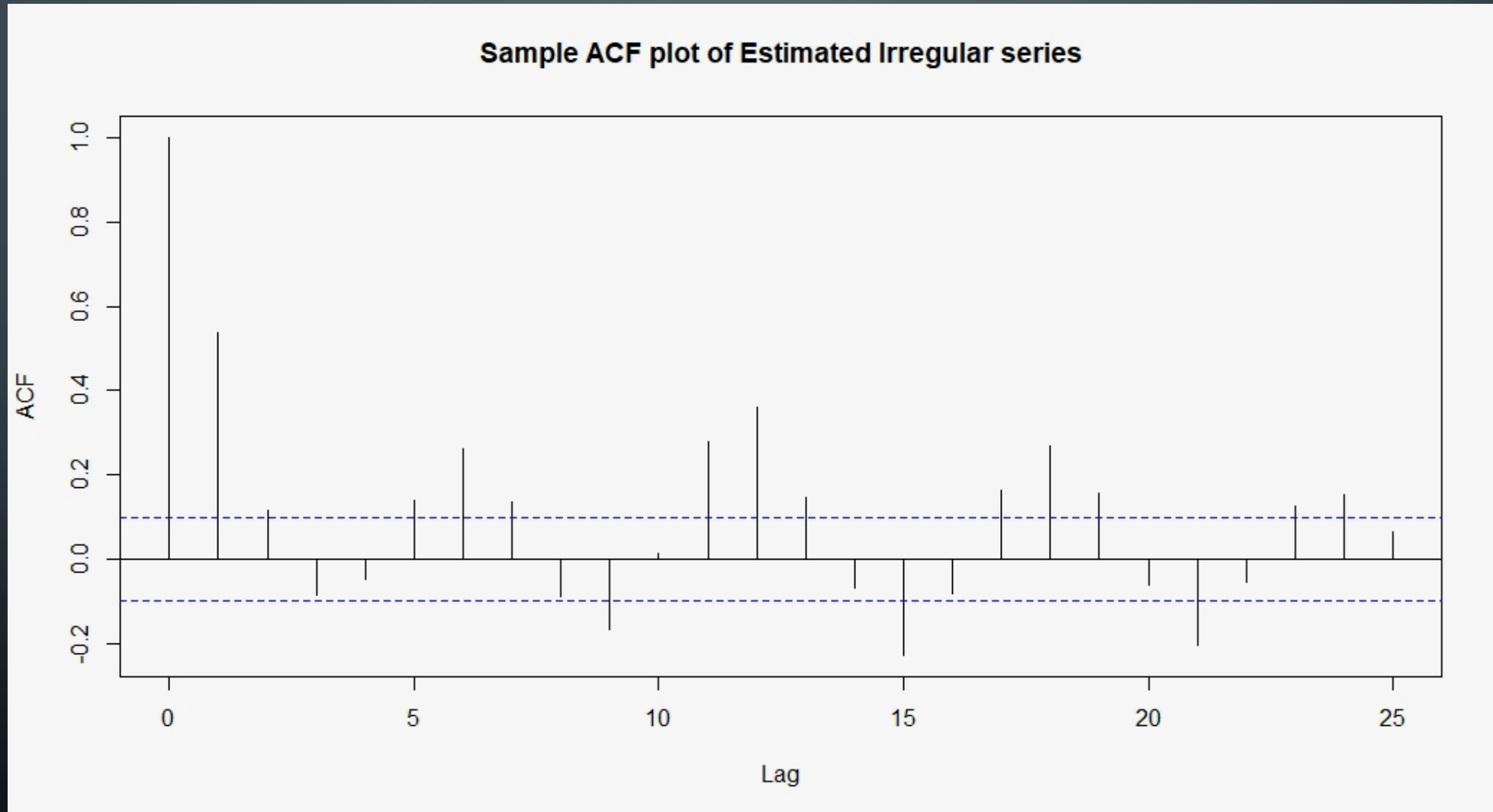
5. ANALYSIS OF THE ESTIMATED IRREGULAR SERIES $\{\hat{I}_t\}$

A. WHITE NOISE CHECKING

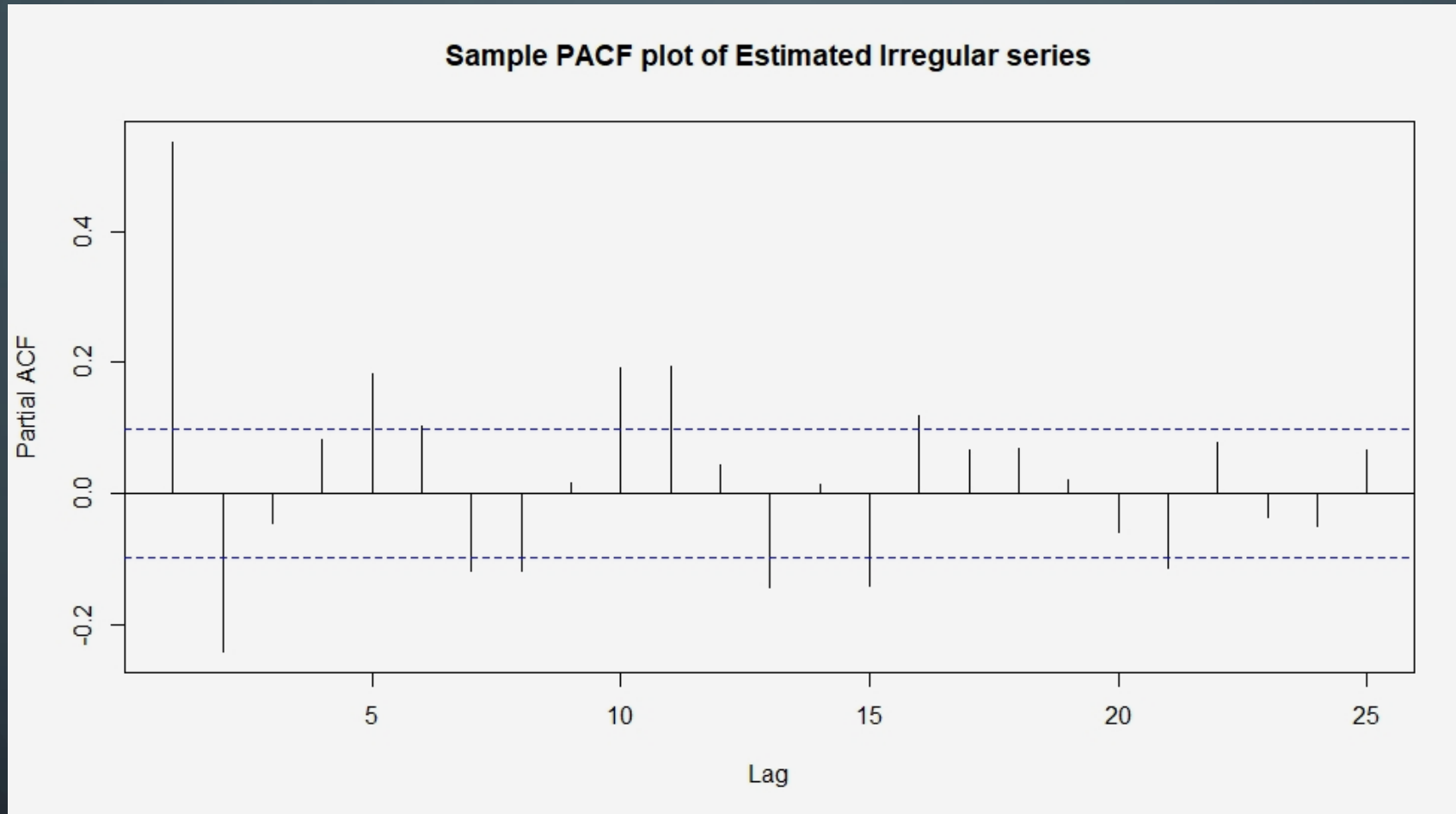
- H_0 : $\hat{I}_t$ is form a white noise process Vs H_1 : not H_0
- Test statistic : $Z = \sqrt{n}\hat{\rho}_1 \sim AN(0,1)$ under H_0
- Rule: Reject H_0 if $|Z_{obs}| > \tau_{\alpha/2}$
- We found $|Z_{obs}| = 10.72 > \tau_{0.025} = 1.96$ hence conclude that $\{\hat{I}_t\}$ not form a white noise process.

B. ACF AND PACF PLOTS

ACF PLOT



PACF PLOT



- We see that in both ACF and PACF plots, the spikes never cut off after a certain lag and they “tail off”.
- So we can suspect that the series follows an $ARMA(p, q)$ process.

C. ESTIMATION OF THE ORDER OF ARMA MODEL

- First we calculate AIC for various p and q's
- Then choose that values for which AIC is minimum

```
> round(aic,2)
```

p	q	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
0		1990.61	1854.76	1837.39	1838.82	1838.73	1834.85	1836.72	1827.35	1824.30	1826.03	1826.38	1826.28	1804.08	1804.21	1800.63
1		1857.24	1840.53	1839.12	1840.43	1839.72	1836.83	1831.13	1826.95	1826.19	1827.34	1828.22	1820.89	1803.63	1805.62	1802.43
2		1834.29	1835.88	1834.24	1794.28	1784.17	1771.38	1764.39	1766.33	1768.28	1770.28	1772.28	1773.79	1775.46	1804.50	1765.71
3		1835.56	1837.25	1766.19	1768.05	1789.84	1784.98	1764.21	1763.40	1764.70	1765.35	1766.65	1767.51	1769.26	1770.76	1766.62
4		1835.09	1829.14	1768.06	1768.08	1770.07	1771.43	1763.22	1764.59	1765.52	1767.16	1765.91	1769.26	1770.66	1771.90	1771.53
5		1823.27	1823.51	1769.92	1770.07	1768.51	1759.15	1772.87	1774.28	1774.87	1768.67	1767.87	1763.29	1771.08	1773.70	1770.83
6		1820.66	1820.60	1767.68	1771.30	1766.75	1770.75	1774.81	1762.54	1761.71	1765.32	1763.06	1769.87	1755.51	1757.11	1757.19
7		1816.45	1816.05	1768.72	1765.21	1767.21	1775.28	1761.67	1763.52	1764.97	1758.80	1765.06	1772.19	1757.05	1759.07	1756.23
8		1812.42	1814.41	1768.04	1767.21	1769.22	1770.02	1763.49	1775.33	1776.01	1763.55	1765.33	1771.25	1756.33	1759.74	1758.22
9		1814.39	1809.02	1768.24	1768.54	1768.80	1767.09	1765.07	1773.50	1775.24	1766.53	1762.92	1760.76	1752.50	1754.39	1756.55
10		1799.60	1792.51	1798.29	1765.36	1765.58	1764.02	1758.52	1767.88	1768.21	1769.03	1764.56	1759.90	1752.10	1753.54	1755.70
11		1787.50	1789.04	1769.53	1770.70	1761.25	1759.12	1759.58	1761.43	1763.38	1764.76	1754.55	1751.33	1752.32	1754.20	1753.37
12		1788.31	1777.88	1777.80	1781.09	1765.79	1754.04	1756.00	1754.97	1758.66	1750.74	1751.03	1751.05	1752.88	1755.35	1755.01
13		1781.55	1777.57	1777.85	1768.48	1767.74	1756.42	1756.44	1757.46	1760.48	1760.89	1751.19	1752.86	1744.51	1745.71	1747.55
14		1783.54	1779.17	1779.65	1766.18	1764.17	1757.70	1757.17	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN

We see that AIC is minimum for $p=13$ and $q=12$. So $\{\hat{I}_t\}$ is form a $ARMA(13,12)$ process.

D. ESTIMATION OF MODEL AND NOISE PARAMETER

- Since estimated irregular component $\hat{I}_t$ is form ARMA($p=13$, $q=12$), so we have to estimate 13 AR and 12 MA parameters, an intercept term and error variance.
- Using MLE we got the following results

```
> model
```

```
Call:
```

```
arima(x = It, order = c(p, 0, q), method = "ML")
```

```
Coefficients:
```

	ar1	ar2	ar3	ar4	ar5	ar6	ar7	ar8
	-0.0627	0.3720	-0.4610	-0.2932	0.1624	0.0568	0.0639	0.4145
s.e.	0.4413	0.2877	0.0845	0.2494	0.1340	0.0941	0.1100	0.0881
	ar9	ar10	ar11	ar12	ar13	ma1	ma2	ma3
	-0.0302	-0.1529	0.6665	0.3734	-0.3808	0.6340	-0.0511	0.4701
s.e.	0.2402	0.1723	0.0699	0.3052	0.1270	0.4513	0.0923	0.0780
	ma4	ma5	ma6	ma7	ma8	ma9	ma10	ma11
	0.6130	0.1587	0.1553	0.1133	-0.5817	-0.3384	0.1771	-0.6187
s.e.	0.3087	0.0970	0.1160	0.0993	0.1122	0.3025	0.1023	0.0916
	ma12	intercept						
	-0.6667	0.0501						
s.e.	0.3716	0.3802						

```
sigma^2 estimated as 4.017: log likelihood = -845.25, aic = 1744.51
```

E. FINAL ESTIMATED MODEL

$$\begin{aligned} Y_t = & 0.0501 - 0.0627Y_{t-1} + 0.3720Y_{t-2} - 0.4610Y_{t-3} - 0.2932Y_{t-4} + 0.1624Y_{t-5} \\ & + 0.0568Y_{t-6} + 0.0639Y_{t-7} + 0.4145Y_{t-8} - 0.0302Y_{t-9} - 0.1529Y_{t-10} + 0.9995Y_{t-11} \\ & + 0.3734Y_{t-12} - 0.3808Y_{t-13} + \epsilon_t + 0.6340\epsilon_{t-1} - 0.0511\epsilon_{t-2} + 0.4701\epsilon_{t-3} \\ & + 0.6130\epsilon_{t-4} + 0.1587\epsilon_{t-5} + 0.1553\epsilon_{t-6} + 0.1133\epsilon_{t-7} + 0.5817\epsilon_{t-8} - 0.3384\epsilon_{t-9} \\ & + 0.1771\epsilon_{t-10} - 0.6187\epsilon_{t-11} - 0.6667\epsilon_{t-12} \end{aligned}$$

6. FORECASTING

- We used Best Linear Predictor (BLP) approach
- Given $X_1, X_2, \dots, X_n$, we want to predict X_{n+h}
- BLP function:

$$P_n(X_{n+h}) = a_0 + \sum_{i=1}^n a_i X_{n+1-i}$$

- a_i 's are such that $E(X_{n+h} - P_n(X_{n+h}))^2$ is minimum

FORECASTING RESULTS

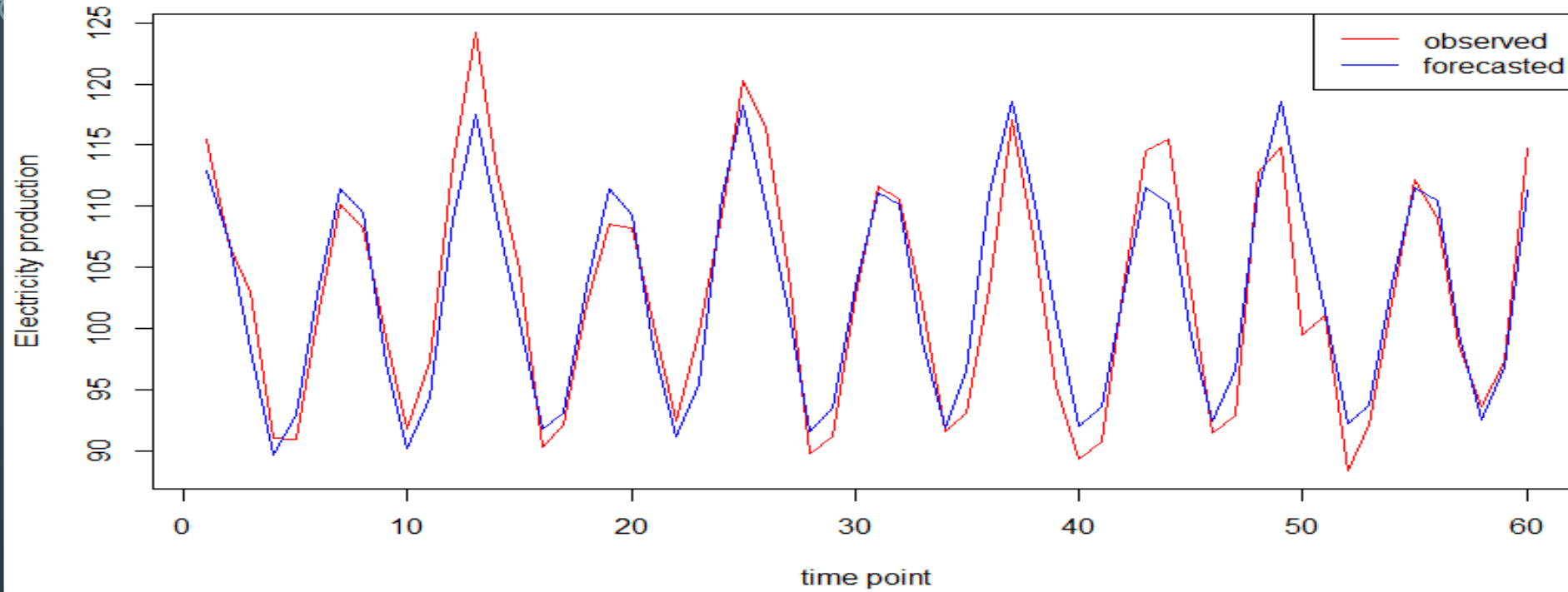
Here are the forecasting results of next 8 years started from Jan-2018 up-to Dec-2025

	Date	Forecasted Values
1	Jan-18	119.6905445
2	Feb-18	111.3616241
3	Mar-18	103.3211018
4	Apr-18	91.16328408
5	May-18	94.02891991
6	Jun-18	101.9815141
7	Jul-18	111.8442562
8	Aug-18	110.1100329
9	Sep-18	99.29552906
10	Oct-18	92.98268902
11	Nov-18	97.20849435
12	Dec-18	110.9591687
13	Jan-19	118.3788477
14	Feb-19	110.0728925
15	Mar-19	101.0413144
16	Apr-19	91.73076104
17	May-19	92.62818057
18	Jun-19	103.1971604
19	Jul-19	112.253745
20	Aug-19	110.1296508

21	Sep-19	100.4181815
22	Oct-19	92.71519147
23	Nov-19	96.34589729
24	Dec-19	110.2846427
25	Jan-20	117.7468115
26	Feb-20	109.4144732
27	Mar-20	101.7050778
28	Apr-20	91.7113755
29	May-20	93.56619043
...	...	...
...	...	...
87	Mar-25	100.2341
88	Apr-25	90.49451
89	May-25	91.36461
90	Jun-25	100.6909
91	Jul-25	108.7323
92	Aug-25	108.0507
93	Sep-25	97.72463
94	Oct-25	90.64823
95	Nov-25	94.42751
96	Dec-25	108.2961

7. JUDGING THE ACCURACY OF OUR METHOD

- To check whether our model is good enough or not for the sake of prediction or forecasting, we cross checked the model for our data dividing it into 2 parts - train set and test set.
- The train set contains the first 336 data points and the test set contains the rest 60 data points
- On repeating the same process on the train set, the model having minimum AIC value turns out to be ARMA(13,12).



- $MSE = 10.51341$
- $MAPE = 2.502531$

Which shows that our model is good enough to predict the electricity production in future.

The background is a dark blue gradient. In the corners, there are white, stylized circuit-like lines with small circles at the ends, resembling a network or data flow diagram.

THANK YOU